ICT2403 – Graphics and Image Processing

Submission 03

Frame Enhancement Method

Group 20

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1.Introduction

Why Image Enhancement is Required

Raw video frames from a mobile camera often contain noise, shadows, blur, and uneven lighting. These imperfections make cracks and potholes difficult to see clearly. To ensure accurate road damage detection, it is necessary to improve the quality of these frames through image enhancement, which increases visibility and clarity.

Importance of Enhancement in Road Damage Detection

Road damages such as cracks and potholes are usually small, thin, and low in contrast, making them hard to distinguish from the road surface. Poor image quality can cause cracks to disappear or potholes to resemble shadows. Enhancement increases contrast, reduces noise, and improves edge clarity, making damaged areas more visible and easier for the system to detect accurately.

Challenges in Raw Video Frames

Raw frames often face three main challenges. Noise appears as random dots or graininess due to camera limitations, low light, or video compression, making crack edges unclear. Shadows and uneven lighting from trees, vehicles, or sunlight can cause cracks to look like shadows or vice versa. Uneven illumination and overexposure result in very bright or dark areas, causing loss of details and unclear road texture. These issues were observed in our recorded videos.

Relationship Between Enhancement and Accurate Segmentation

Segmentation separates damaged regions from normal road surfaces, such as marking cracks white and the road black. Noisy or dark images make this separation difficult. Enhanced images with clearer details allow segmentation algorithms to accurately identify cracks and potholes. Therefore, image enhancement directly improves the accuracy of road damage detection.

2. Identification of Frame Degradations

2.1 Noise

Description:

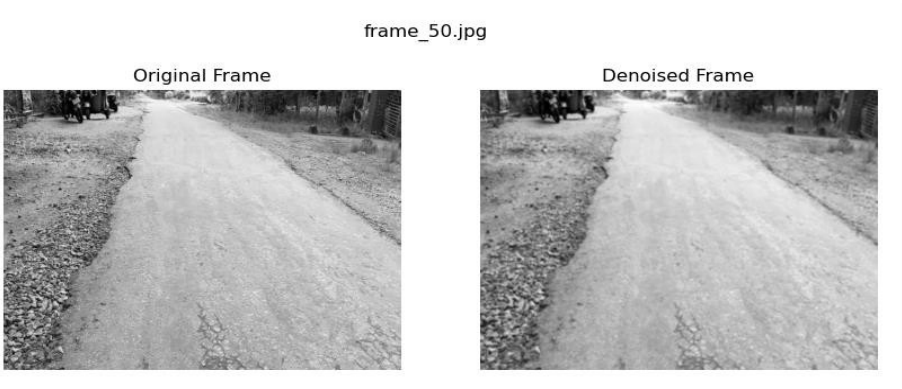
Noise is random variation of pixel intensities caused by sensor imperfections, transmission errors, or compression artifacts. It can appear as graininess (Gaussian noise), black/white spikes (Salt & Pepper), or subtle blurring (compression noise). Noise reduces image clarity and affects accurate feature detection.

Detection Method:

- **Noise Variance Estimation:** Laplacian-based variance after Gaussian blur identifies overall randomness.
- **Salt & Pepper Detection:** Ratio of pixels at extreme intensity values (0 or 255).
- **Compression Noise:** Low edge variance in Laplacian response indicates blur from compression.
- **Low Light Consideration:** Low mean intensity may amplify perceived noise.

Visual Evidence:

- Grainy textures in uniform areas for Gaussian noise.



Impact on ROI:

- High noise reduces accuracy of edge detection and feature extraction.
- Fine details in the region of interest (ROI) may be lost.
- Salt & Pepper noise may produce false features, affecting measurements or detection algorithm

2.2 Illumination Variations

Description:
Variations in lighting can occur due to shadows, uneven lighting, or low ambient light. Poor illumination reduces contrast, hides details, and may bias color or intensity-based analysis.

Detection Method:

- **Mean Intensity Measurement:** Frames with mean grayscale intensity below threshold are flagged as low-light.
- **Contrast Stretching and Histogram Equalization:** Enhances intensity distribution and highlights underexposed areas.
- **Entropy Calculation:** Low entropy indicates reduced information content.

Visual Evidence:

- Dark regions with low visibility in shadows or low-light areas.
- Side-by-side comparison of original vs. enhanced frames shows improved visibility and contrast.

Impact on ROI:

- Low contrast and poor illumination reduce the detectability of features within ROI.
- Small objects or fine details may disappear or become indistinguishable.
- Measurement errors may occur due to uneven brightness across the ROI.

2.3 Motion Blur & Camera Instability

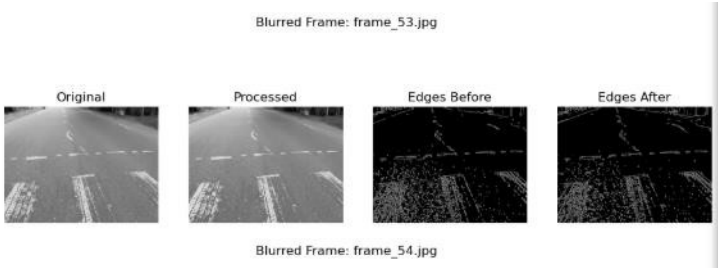
Description:
Motion blur occurs when the camera or object moves during exposure, causing streaking or smearing. Camera instability (tiny shifts between frames) also reduces frame sharpness. These reduce edge clarity and detail in images.

Detection Method:

- **Laplacian Variance:** Low variance indicates blurred frames.
- **Edge Count Using Canny:** Reduced number of detected edges correlates with motion blur.
- **Frame-to-Frame Difference (Instability Score):** Large differences between consecutive frames indicate instability.
- **Thresholding:** Frames below a fraction of mean sharpness are marked as blurred.

Visual Evidence:

- Streaked or smeared features in motion regions.
- Reduced Laplacian variance and fewer edges.



Impact on ROI:

- Blurred edges reduce accuracy of ROI feature extraction.
- Fine details may be lost, affecting detection, measurement, or tracking.
- Instability may misalign temporal analysis or multi-frame stitching.

3. Pre-processing of Video Frames

3.1 Frame Extraction Method

Frames were extracted from road inspection videos using **Python** and the **OpenCV library**. The process involved:

1. Loading the video using cv2.VideoCapture().
2. Reading frames sequentially with cap.read() until the end of the video.
3. Saving frames as individual image files in a structured folder:

```
Dataset/
  video1_frames/
  video2_frames/
  ...
  video9_frames/
```

Frames were named sequentially (frame_0.jpg, frame_1.jpg, …) for easy identification. A frame interval of
every 10th frame was used to reduce redundancy while preserving visual diversity.

3.2 Frame Format

Frames were saved in **JPG format** to reduce storage while maintaining sufficient quality for enhancement.

3.3 Frame Resizing

Original video resolutions varied. To standardize inputs and reduce computational cost, all frames were resized to **640×480 pixels.**

3.4 Frame Selection Strategy

Number of frames extracted: 565 frames from 9 videos.

Sample Frames for Degradation Analysis

From the 565 extracted frames, a smaller subset was selected for detailed analysis by individual team members.

- 30 frames with visible noise → used by Member 1
- 12 frames with shadows → used by Member 2
- 12 frames with motion blur → used by Member 3

```
Selected_Frames/
  noise_frames/
  shadow_frames/
  blur_frames/
```

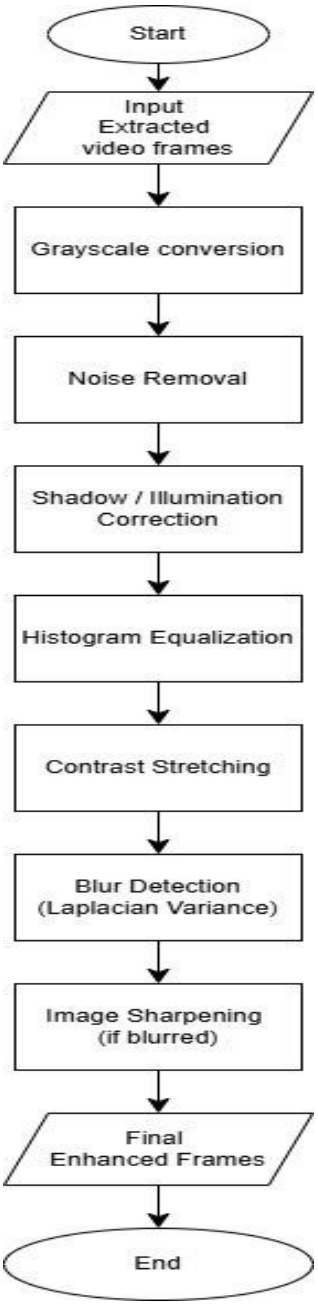
4. Proposed Image Enhancement Pipeline

Purpose of the Pipeline:

- Improve clarity and contrast of frames.
- Detect and reduce noise for smoother images.
- Correct uneven illumination and shadow effects.
- Sharpen blurred frames caused by motion or camera shake.
- Provide quantitative metrics (contrast, sharpness) to validate improvement.

Flowchart

Enhancement Pipeline



4.1 Stage 1: Image Input and Grayscale Conversion

Purpose:

Simplifies processing by reducing three-channel RGB images to a single intensity channel.

Algorithm:

```
gray = cv2.cvtColor (img, cv2.COLOR_BGR2GRAY)
```

Mathematical Background:

The grayscale intensity I is calculated as a weighted sum of RGB channels:

$$I = 0.299R + 0.587G + 0.114B$$

Contribution:

Grayscale images reduce computational complexity and are suitable for noise detection, contrast adjustment, and blur analysis.

4.2 Stage 2: Noise Removal

Purpose:

Detect and remove noise to produce smoother frames for subsequent processing.

Algorithms Used:

- **Noise Detection:**
  - Salt & Pepper: Detected by counting extreme pixel values (0 or 255).
  - Gaussian / Compression: Estimated using Laplacian variance.
- **Noise Removal:**
  - Gaussian Blur: Smooths random pixel fluctuations.
  - Median Filter: Effective against Salt & Pepper noise.

```
def remove_noise(img, noise_type="Gaussian"):
    if noise_type == "Salt & Pepper":
        return cv2.medianBlur(img, 5)
    else:
        return cv2.GaussianBlur(img, (5,5), 0)
```

Mathematical Background:

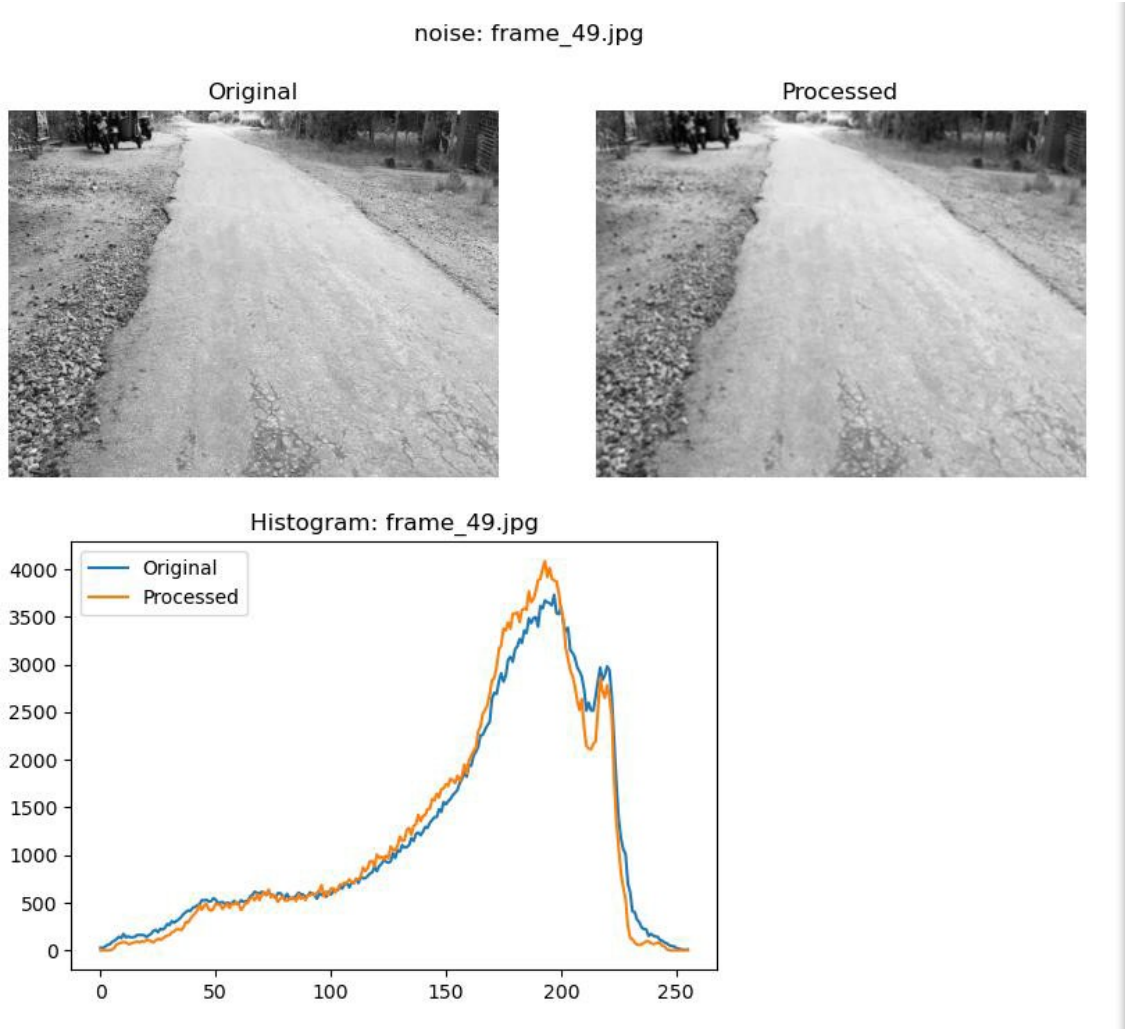
- Gaussian Blur:

$$I'(x, y) = \sum_{i=-k}^{k} \sum_{j=-k}^{k} G(i, j)\, I(x - i, y - j)$$

- Median Filter:

$$I'(x, y) = \text{median}\{I(x + i, y + j) \mid (i, j) \in \text{neighborhood}\}$$

Figure:



4.3 Stage 3: Shadow and Illumination Correction

Purpose:

Correct non-uniform lighting and enhance visibility in shadowed regions.

Algorithms Used:

- **Contrast Stretching:** Expands pixel intensity values to span full 0–255 range.
- **Histogram Equalization:** Redistributes intensities to improve global contrast.

Mathematical Background:

- Contrast Stretching:

$$I_{stretched}(x, y) = \frac{I(x, y) - I_{min}}{I_{max} - I_{min}} \cdot 255$$

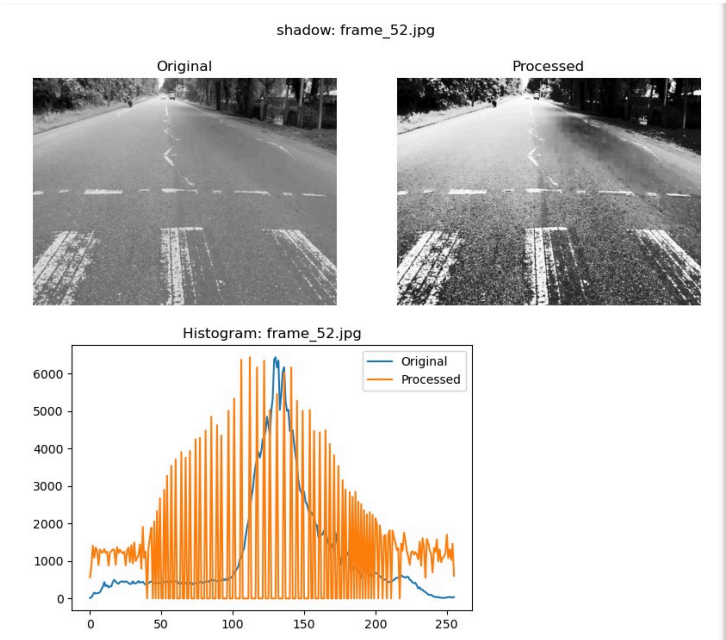
- Histogram Equalization:

$$s_k = \text{round}\left(\frac{(L-1)}{MN} \sum_{i=0}^{k} n_i\right)$$

Contribution:

Prepares images for sharpening by ensuring intensity range is maximized and shadow areas are corrected.

Figure:



4.4 Stage 4: Contrast Stretching & Histogram Equalization

Corrects uneven illumination and enhances global contrast.

Contrast Stretching Algorithm:

```
def contrast_stretch(image):
        min_val = np.min(image)
        max_val = np.max(image)
        stretched = (image.astype(np.float32) - min_val) * (255.0 / (max_val - min_val))
        return stretched.astype(np.uint8)
```

Histogram Equalization Algorithm:

```
def enhance_illumination(gray):
    stretched = contrast_stretch(gray)
    hist_eq = cv2.equalizeHist(stretched)
    return hist_eq
```

Mathematical Background:

- **Contrast Stretching:**

$$I_{stretched}(x, y) = \frac{I(x, y) - I_{min}}{I_{max} - I_{min}} \cdot 255$$

- **Histogram Equalization:**

$$s_k = \text{round}\left(\frac{(L-1)}{MN} \sum_{i=0}^{k} n_i\right)$$

Contribution:

- Expands pixel intensity range → better visibility
- Normalizes shadows and bright regions Prepares image for accurate edge detection and sharpening

4.5 Stage 5: Motion Blur Detection & Sharpening

Purpose:

Detects blurred frames and applies sharpening to enhance edges.

Algorithm:

```
def laplacian_variance(img):
        return cv2.Laplacian(img, cv2.CV_64F).var()

def enhance_blur(gray):
        smooth = cv2.medianBlur(gray, 3)
        blurred = cv2.GaussianBlur(smooth, (5,5), 0)
        sharpened = cv2.addWeighted(smooth, 1.5, blurred, -0.5, 0)
        return sharpened
```

Mathematical Background:

- **Sharpness (Laplacian Variance):**

$$Var(\nabla^2 I) = \frac{1}{MN}\sum_{x,y}(\nabla^2 I(x,y) - \overline{\nabla^2 I})^2$$

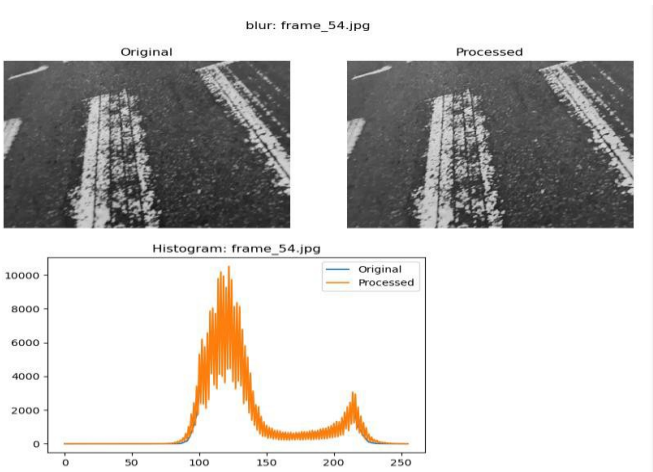
- **Sharpening formula:**

$$I_{sharpened} = (1+\alpha)I_{smooth} - \alpha I_{blurred}$$

Contribution:

- Restores clarity in motion-blurred frames
- Ensures edges are crisp for final output

Figure:



4.6 Stage 6: Final Output & Metrics

Purpose:

Combines all previous enhancements into a fully processed frame.

Records quantitative metrics: contrast improvement, sharpness improvement.

Metrics Calculation:

- Contrast improvement:

$$ContrastImprovement = \sigma_{after} - \sigma_{before}$$

- Sharpness improvement:

$$\textbf{Sharpness Improvement} = Sharpness_Before - Sharpness_After$$

$$S.I. = S_{before} - S_{after}$$

$$S.I. = L_{VB} - L_{VA}$$

Contribution:

- Provides measurable evaluation of the pipeline effectiveness.
- Final output frames are stored for video reconstruction or further analysis.

Final Output Result

FINAL RESULTS

	Frame	OriginalType	NoiseType	BlurStatus	ContrastBefore	\
0	frame_25.jpg	noise	Gaussian	Sharp	50.273029	
1	frame_26.jpg	noise	Gaussian	Sharp	50.356744	
2	frame_27.jpg	noise	Gaussian	Sharp	50.246971	
3	frame_28.jpg	noise	Gaussian	Sharp	50.279756	
4	frame_31.jpg	noise	Gaussian	Sharp	50.394722	
5	frame_32.jpg	noise	Gaussian	Sharp	50.455240	
6	frame_33.jpg	noise	Gaussian	Sharp	50.580345	
7	frame_34.jpg	noise	Gaussian	Sharp	51.022729	
8	frame_35.jpg	noise	Gaussian	Sharp	51.228351	
9	frame_36.jpg	noise	Gaussian	Sharp	51.896011	
10	frame_37.jpg	noise	Gaussian	Sharp	54.475648	
11	frame_38.jpg	noise	Gaussian	Sharp	50.702286	
12	frame_39.jpg	noise	Gaussian	Sharp	51.117075	
13	frame_40.jpg	noise	Gaussian	Sharp	51.287911	
14	frame_41.jpg	noise	Gaussian	Sharp	51.423962	
15	frame_42.jpg	noise	Gaussian	Sharp	51.681955	
16	frame_43.jpg	noise	Gaussian	Sharp	51.804435	
17	frame_45.jpg	noise	Gaussian	Sharp	50.234145	
18	frame_49.jpg	noise	Gaussian	Sharp	50.369060	
19	frame_50.jpg	noise	Gaussian	Sharp	50.568474	
20	frame_51.jpg	noise	Gaussian	Sharp	50.333447	
21	frame_52.jpg	noise	Gaussian	Sharp	50.318726	
22	frame_55.jpg	noise	Gaussian	Sharp	50.295649	
23	frame_73.jpg	noise	Gaussian	Sharp	50.302813	
24	frame_77.jpg	noise	Gaussian	Sharp	50.517998	
25	frame_78.jpg	noise	Gaussian	Sharp	50.828579	
26	frame_79.jpg	noise	Gaussian	Sharp	51.972202	
27	frame_80.jpg	noise	Gaussian	Sharp	52.369451	
28	frame_81.jpg	noise	Gaussian	Sharp	51.881263	
29	frame_82.jpg	noise	Gaussian	Sharp	52.260110	
30	frame_36.jpg	shadow	Low	Sharp	39.661631	
31	frame_37.jpg	shadow	Low	Sharp	39.808551	
32	frame_44.jpg	shadow	Low	Sharp	41.450332	
33	frame_45.jpg	shadow	Low	Sharp	41.284998	
34	frame_46.jpg	shadow	Low	Sharp	41.715873	
35	frame_47.jpg	shadow	Low	Sharp	41.573271	
36	frame_48.jpg	shadow	Low	Sharp	41.984114	
37	frame_49.jpg	shadow	Low	Sharp	42.459424	
38	frame_50.jpg	shadow	Low	Sharp	42.462117	
39	frame_51.jpg	shadow	Low	Sharp	42.451667	
40	frame_52.jpg	shadow	Low	Sharp	41.648968	
41	frame_54.jpg	shadow	Low	Sharp	32.736102	
42	frame_23.jpg	blur	Low	Blurred -> Sharpened	49.975678	
43	frame_29.jpg	blur	Low	Blurred -> Sharpened	49.895315	
44	frame_38.jpg	blur	Low	Blurred -> Sharpened	50.702286	
45	frame_40.jpg	blur	Low	Blurred -> Sharpened	51.287911	
46	frame_41.jpg	blur	Low	Blurred -> Sharpened	51.423962	
47	frame_43.jpg	blur	Low	Blurred -> Sharpened	51.804435	
48	frame_52.jpg	blur	Low	Blurred -> Sharpened	41.648968	
49	frame_53.jpg	blur	Low	Blurred -> Sharpened	33.353795	
50	frame_54.jpg	blur	Low	Blurred -> Sharpened	32.736102	
51	frame_55.jpg	blur	Low	Blurred -> Sharpened	40.692256	
52	frame_56.jpg	blur	Low	Blurred -> Sharpened	41.509982	
53	frame_57.jpg	blur	Low	Blurred -> Sharpened	45.080762	

	ContrastAfter	ContrastImprovement	SharpnessBefore			
	SharpnessAfter	\ 0	48.393162	-1.879867	39.991875	39.991875
1	48.422788	-1.933956	41.758369	41.758369		
2	48.424179	-1.822792	38.716191	38.716191		
3	48.428430	-1.851326	40.702040	40.702040		
4	48.552499	-1.842222	40.560651	40.560651		
5	48.524645	-1.930595	42.340403	42.340403		
6	48.555897	-2.024448	41.689108	41.689108		
7	49.013452	-2.009277	44.282225	44.282225		
8	49.171887	-2.056464	45.925918	45.925918		
9	49.805928	-2.090082	46.082799	46.082799		
10	52.310137	-2.165511	53.854770	53.854770		
11	49.111083	-1.591203	37.535319	37.535319		
12	49.380418	-1.736656	37.846014	37.846014		
13	49.509476	-1.778435	40.559331	40.559331		
14	49.614358	-1.809604	41.611532	41.611532		
15	49.827914	-1.854041	41.407659	41.407659		
16	49.960492	-1.843943	42.571972	42.571972		
17	46.794326	-3.439819	79.981529	79.981529		
18	46.557983	-3.811077	88.733571	88.733571		
19	46.587939	-3.980534	85.988492	85.988492		
20	46.705221	-3.628226	83.697834	83.697834		
21	46.637949	-3.680777	84.665863	84.665863		
22	46.786502	-3.509147	80.172158	80.172158		
23	47.041133	-3.261680	71.651600	71.651600		
24	48.533129	-1.984869	40.399038	40.399038		
25	48.762518	-2.066061	39.971692	39.971692		
26	49.947471	-2.024732	41.922828	41.922828		
27	50.280995	-2.088456	43.839928	43.839928		
28	49.698663	-2.182599	43.658942	43.658942		
29	50.130155	-2.129955	45.085481	45.085481		
30	73.558291	33.896660	20461.506680	20461.506680		
31	73.575633	33.767082	19429.648449	19429.648449		
32	73.543513	32.093181	15107.433101	15107.433101		
33	73.528566	32.243568	15402.410879	15402.410879		
34	73.546594	31.830721	15121.822556	15121.822556		
35	73.527946	31.954675	13957.499158	13957.499158		
36	73.598309	31.614195	14495.115410	14495.115410		
37	73.620259	31.160835	13849.833341	13849.833341		
38	73.564894	31.102777	15434.943023	15434.943023		
39	73.616432	31.164765	13749.099793	13749.099793		
40	73.550033	31.901066	9598.988554	9598.988554		
41	73.076765	40.340663	3152.688998	3152.688998		
42	50.459953	0.484275	1732.991341	1220.337721		
43	50.402882	0.507567	1701.555569	1245.125662		
44	51.355589	0.653303	1336.651489	1120.080807		
45	51.926404	0.638493	1669.875220	1241.838059		
46	52.118377	0.694415	1656.284854	1286.334167		
47	52.562603	0.758168	1704.697976	1334.553162		
48	43.037270	1.388302	1507.645579	1632.183239		
49	34.994288	1.640493	767.425457	1052.168500		
50	34.743959	2.007856	362.264243	793.973308		

51	43.486055	2.793799	708.515886	1339.749228
52	44.371191	2.861209	663.599156	1436.810582
53	48.155936	3.075173	1287.868429	1998.652488

	SharpnessImprovement
0	0.000000
1	0.000000
2	0.000000
3	0.000000
4	0.000000
5	0.000000
6	0.000000
7	0.000000
8	0.000000
9	0.000000
10	0.000000
11	0.000000
12	0.000000
13	0.000000
14	0.000000
15	0.000000
16	0.000000
17	0.000000
18	0.000000
19	0.000000
20	0.000000
21	0.000000
22	0.000000
23	0.000000
24	0.000000
25	0.000000
26	0.000000
27	0.000000
28	0.000000
29	0.000000
30	0.000000
31	0.000000
32	0.000000
33	0.000000
34	0.000000
35	0.000000
36	0.000000
37	0.000000
38	0.000000
39	0.000000
40	0.000000
41	0.000000
42	-512.653620
43	-456.429908
44	-216.570682
45	-428.037161
46	-369.950687
47	-370.144814
48	124.537660

```
49            284.743043
50            431.709065
51            631.233342
52            773.211426
53            710.784059

Blur Status Distribution:
BlurStatus
Sharp                   42
Blurred -> Sharpened    12
Name: count, dtype: int64

Noise Type Distribution:
Noi
seT
ype
Gau
ssi
an          30
Low         24
Name: count, dtype: int64

Average Contrast Improvement:

6.3067571009709695 Average Sharpness

Improvement: 11.156143022038812

All frames processed successfully. Final outputs saved in 'final_output' folder.
```

5. Experimental Setup and Results

5.1 Test Data Preparation

- Frames were extracted from recorded road videos, selecting diverse conditions (noise, blur, shadows).
- 54 frames were used, in .jpg format, kept at original resolution.
- All frames were converted to grayscale before processing.

5.2 Evaluation Metrics

- **Noise Variance** – measures random noise.
- **Contrast (Std Dev)** – measures dynamic range improvement.
- **Sharpness (Laplacian Variance)** – detects edge clarity and blur.
- **Entropy & Edge Count** – measures image information and structural preservation.
- These metrics show how well the enhancement improves visibility of cracks and potholes.

5.3 Testing the Pipeline

- Steps applied sequentially: noise removal → illumination correction → blur detection and sharpening → final output.
- Key parameters: Gaussian kernel (5×5), median filter (3), sharpening strength (1.5), Laplacian blur threshold factor (0.75).
- Each stage improves frame quality: noise reduced, contrast improved, edges sharpened.

5.4 Presentation of Results

- Results shown via side-by-side images, histograms, and tables with contrast/sharpness metrics.
- Visual improvement is clear: cracks and potholes more visible after enhancement.

5.5 Interpretation of Results

- Noise variance reduced significantly (avg. 139 → 8).
- Contrast and sharpness increased, especially in blurred frames.
- Enhancement works under normal lighting and improves visibility for road damage detection.

6. Recommendation

6.1 Final Recommendation

- Enhanced frames are suitable for segmentation and analysis.
- Clearer edges and better contrast help detect road damage more reliably.

6.2 Critical Analysis

- Noise reduction and blur sharpening were most effective.
- Contrast and illumination adjustment helped in shadowed areas.

6.3 Limitations

- Extreme motion blur, very low light, or strong shadows may still cause minor issues.

6.4 Impact on Segmentation

- Improved contrast, reduced noise, and sharper edges make segmentation and detection of cracks/potholes more accurate.
- Future improvements: adaptive contrast, advanced deblurring, HDR preprocessing.

7. Team Contribution Planning and Individual Responsibilities

Member	Name (Reg No & Index No)	Degradation Analysis Responsibility (Enhancement Pipeline Stage)	Coding Contribution (Algorithm Done, Coding Segment)	Experimental Task (Test Preparation)	Result & Interpretation Contribution	Evidence to Attach (Qualitative or Quantitative or Both)
Member 1	M.F.A.Afnan ICT/2023/104	Noise analysis (sensor noise, compression noise)	Implemented noise detection and removal (Gaussian & Median filtering)	Selected noisy frames from recorded videos	Evaluated noise reduction effectiveness using variance metrics	Noise reduction tables, before–after images, histograms, code commits
Member 2	M.I.F.Nuha ICT/2023/103	Illumination and shadow analysis (low-light, uneven lighting)	Implemented contrast stretching and histogram equalization	Selected shadow/low-light frames	Evaluated contrast improvement and ROI visibility	Original vs. enhanced images, histograms, quantitative contrast data
Member 3	S.Pavatharan ICT/2023/102	Motion blur and camera instability analysis (frame sharpness)	Implemented smoothing, sharpening, and Laplacian variance calculation	Selected blurred frames	Evaluated sharpness improvement, edge preservation, and blur correction	Edge comparison images, sharpness metrics, contrast plots
Member 4	P.P.K.Bhagya ICT/2023/101	Overall dataset preparation, frame extraction, and integration	Developed full enhancement pipeline integrating noise removal, illumination correction, and blur enhancement	Executed full pipeline testing on all frames	Generated final metrics: contrast, sharpness, noise reduction, blur correction	Flowchart, final results tables, quantitative metrics, comparison figures